

A mathematical modelling of rheumatoid arthritis

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We model the interaction between three populations of cells in the context of arthritis, namely the neutrophils, the tissue resident macrophages (SubLining Tissue Resident Macrophages) and the Monocyte Macrophages. The SL-TRM macrophage cells reside in the outer border of the joint. The Momac (monocytes-macrophages) cells are coming from the reservoir of monocytes in the marrow. Basically the crisis scenario is the following: The arthritis crisis is triggered by an external signal (injection of immune serum containing antigens) that activates the immune system and provokes an inflammatory response. The antigens are recognized by the macrophages L-TRM (lining tissue resident macrophages) cells that secrete inflammatory chemokines and cytokines. These cytokines activate the fibroblasts which in turn recruit the neutrophils cells from the blood stream into the joint. The cytokines also recruit the Momac cells from the blood stream into the joint. There is a positive feedback since the inflammation is amplified by the neutrophils that are recruited in the joint by the chemokines secreted by the Momac cells. The macrophages SL-TRM cells have a protective role since they inhibit the inflammation but they are depleted during the crisis because they are penalized by the proliferation of Momac cells.

1 Model

Basically the crisis scenario is the following: The arthritis crisis is triggered by an external signal (injection of immune serum containing antigens) that activates the immune system and provokes an inflammatory response. The antigens are recognized by the macrophages L-TRM (lining tissue resident macrophages) cells that secrete inflammatory chemokines and cytokines. These cytokines activate the fibroblasts which in turn recruit the neutrophils cells from the blood stream into the joint. The cytokines also recruit the Momac cells from the blood stream into the joint. There is a positive feedback since the inflammation is amplified by the neutrophils that are recruited in the joint by the chemokines secreted by the Momac cells. The macrophages SL-TRM cells have a protective role since they inhibit the inflammation but they are depleted during the crisis because they are inhibited by the proliferation of Momac cells.

1.1 Mathematical Model

Let us denote by $T(t)$ the population of SL-TRM cells and by $M(t)$ the population of Momac cells at time t . The cells T (SL-TRM) and M (Momac) interact with each other in the following way. The Momac cells M are recruited from the blood stream at a constant rate a . They die at a rate δ_M and they transform into SL-TRM cells at a rate b . The SL-TRM cells T die at a rate δ_T . The interaction between the two populations is described by the following system of differential equations

The inflammatory state $S(t)$ is enhanced by neutrophils N and inhibited by SL-TRM macrophages T .

Inflammation further increases neutrophils (factor α), increases Momac (factor β), and decreases SL-TRM (factor γ).

Notice that we do not keep track of the population of fibroblasts, but rather the population of neutrophils, which are recruited by inflammatory chemokines.

The variable $N(t)$ designates the population of neutrophils at time t . Finally $S(t)$ is the inflammatory signal at time t . The model is given by the following system of ordinary differential equations:

$$\begin{cases} \frac{dS}{dt} = (a N - b T)S \\ \frac{dN}{dt} = \alpha S N - \delta_N N(N - 1) \\ \frac{dM}{dt} = \beta S - \delta_M M(M - 1) \\ \frac{dT}{dt} = -\gamma S T + \delta_M M(1 - T) \end{cases} \quad (1)$$

where $a, b, \alpha, \beta, \gamma, \delta_N, \delta_M$ are positive parameters. To make things easier to understand, we summarize the model in the following diagram (see Figure 1.1).

Remark (Comments and questions on the model):

- to be discussed with our immunologists collaborators

2 Analysis of the model

2.1 Equilibria points

Equilibria points of the system are found by setting the right-hand side of the system to zero, i.e. by solving the following system of equations: 01 The equilibrium points (S^*, N^*, M^*, T^*) satisfy the following system:

$$(aN^* - bT^*)S^* = 0 \quad (2)$$

$$\alpha S^* N^* - \delta_N N^*(N^* - 1) = 0 \quad (3)$$

$$\beta S^* - \delta_M M^*(M^* - 1) = 0 \quad (4)$$

$$-\gamma S^* T^* + \delta_M M^*(1 - T^*) = 0 \quad (5)$$

This gives the following equilibria points. A first natural equilibrium point is given by $S^* = 0, N^* = 1, M^* = 1, T^* = 1$. This point corresponds to the disease-free state, where there is no inflammation ($S^* = 0$) and the populations of neutrophils, Momac,



Figure 1: Schematic representation of the model

and SL-TRM are at their baseline levels ($N^* = 1, M^* = 1, T^* = 1$). There are other equilibria points with $S^* = 0$ namely $S^* = 0, N^* = 0, M^* = 0, T^* = R$, with arbitrary R ; $S^* = 0, N^* = 0, M^* = 1, T^* = 1$; $S^* = 0, N^* = 1, M^* = 0, T^* = R$, with arbitrary R .

For equilibria points with $S^* \neq 0$, from equation (2) we have $aN^* - bT^* = 0$, hence

$$T^* = \frac{a}{b}N^*. \quad (6)$$

Factoring N^* from equation (3), we obtain:

$$N^* (\alpha S^* - \delta_N(N^* - 1)) = 0$$

This gives us two possibilities: either $N^* = 0$ or $\alpha S^* - \delta_N(N^* - 1) = 0$. If $N^* = 0$, then from equation (2) we have $a \cdot 0 - bT^* = 0$, hence $T^* = 0$. From equation (5), we get $M^* = 0$. Then from equation (4), we have $\beta S^* - \delta_M \cdot 0 = 0$, hence $S^* = 0$. This contradicts our assumption that $S^* \neq 0$. Therefore, the case $N^* = 0$ is not possible for equilibria with $S^* \neq 0$. Therefore $\alpha S^* - \delta_N(N^* - 1) = 0$, so we can express S^* in terms of N^* :

$$S^* = \frac{\delta_N}{\alpha}(N^* - 1) \quad (7)$$

We substitute (6)-(7) into equation (4) and (5) and obtain a 2 by 2 system for M^* and N^* : By substituting $T^* = \frac{a}{b}N^*$ and $S^* = \frac{\delta_N}{\alpha}(N^* - 1)$ into the equilibrium equations for

M^* and T^* , we obtain the following system for N^* and M^* :

$$\begin{cases} \beta \frac{\delta_N}{\alpha} (N^* - 1) - \delta_M M^* (M^* - 1) = 0 \\ -\gamma \frac{\delta_N}{\alpha} (N^* - 1) \cdot \frac{a}{b} N^* + \delta_M M^* \left(1 - \frac{a}{b} N^*\right) = 0 \end{cases}$$

This system allows us to determine the equilibrium values N^* and M^* when $S^* \neq 0$. Indeed we can express N^* in terms of M^* from the first equation:

$$N^* = 1 + \frac{\alpha \delta_M}{\beta \delta_N} M^* (M^* - 1) \quad (8)$$

Substituting this expression for N^* into the second equation, we obtain an equation for M^* :

$$M^* = \frac{a \gamma \delta_N}{b \alpha \delta_M} \frac{N^* (N^* - 1)}{(1 - \frac{a}{b} N^*)} \quad (9)$$

Geometrically, the equilibria points with $S^* \neq 0$ correspond to the intersection points of the parabola defined by equations (8) and the hyperbola defined by equation (9) in the (N^*, M^*) plane. The number of such intersection points (and hence the number of equilibria with $S^* \neq 0$) depends on the parameters of the model. We can find the asymptotes of the hyperbola and the apex of the parabola to determine the possible number of intersection points in the first quadrant. We rewrite the equation (9) of the hyperbola as follows:

$$M^* = -\frac{\gamma \delta_N}{\alpha \delta_M} (N^* + b/a - 1) + \frac{(b/a)^2 (1 - b/a)}{(N^* - b/a)} \quad (10)$$

From this expression, we see that the hyperbola has a vertical asymptote at $N^* = b/a$ and an oblique asymptote $M^* = -\frac{\gamma \delta_N}{\alpha \delta_M} (N^* + b/a - 1)$. The parabola (8) is horizontal in the N, M plane, has its apex at $N_{min} = 1 - \frac{\alpha \delta_M}{4\beta \delta_N}$ and $M = 1/2$ and is symmetric with respect to the horizontal line $M = 1/2$ and opens eastwards. We plot some typical cases in the following figure. Notice that we need only 3 *non dimensional* parameters a/b , α/β , $\frac{\gamma \delta_N}{\alpha \delta_M}$ instead of the 7 original parameters.

There is a smarter way¹ to find the number of equilibria points with $S^* \neq 0$ leading to a cubic equation in T^* . We express N^* in terms of T^* from equation (6):

$$N^* = \frac{b}{a} T^* \quad (11)$$

Substituting this expression for N^* into equation (7), we obtain:

$$S^* = \frac{\delta_N}{\alpha} \left(\frac{b}{a} T^* - 1 \right) \quad (12)$$

From equation (4) and equation (5), we derive $\delta_M M^* = \frac{\beta S^*}{M^* - 1}$ and $\delta_M M^* = \frac{\gamma S^* T^*}{1 - T^*}$. Equating these two expressions for $\delta_M M^*$, we obtain:

$$\frac{\beta S^*}{M^* - 1} = \frac{\gamma S^* T^*}{1 - T^*}$$

¹thanks to O. Iosifescu



Figure 2: equilibria in the (N^*, M^*) plane, left $a < b$, right $a > b$.

Assuming $S^* \neq 0$, we can simplify this equation to:

$$\frac{\beta}{M^* - 1} = \frac{\gamma T^*}{1 - T^*}$$

From this, we can express M^* in terms of T^* :

$$M^* = 1 + \frac{\beta(1 - T^*)}{\gamma T^*} = 1 - \frac{\beta}{\gamma} + \frac{\beta}{\gamma T^*} \quad (13)$$

Substituting the expression for M^* and S^* into the last equation (5), we obtain

$$-\gamma \frac{\delta_N}{\alpha} \left(\frac{b}{a} T^* - 1 \right) T^* + \delta_M \left(1 - \frac{\beta}{\gamma} + \frac{\beta}{\gamma T^*} \right) (1 - T^*) = 0 \quad (14)$$

Since $T^* \neq 0$, we can multiply this equation by T^* and rearrange the terms to obtain a cubic equation for T^* :

$$\frac{\gamma \delta_N b}{\alpha \delta_M a} (T^*)^3 + \left(1 - \frac{\gamma \delta_N}{\alpha \delta_M} - \frac{\beta}{\gamma} \right) (T^*)^2 + \left(2 \frac{\beta}{\gamma} - 1 \right) T^* - \frac{\beta}{\gamma} = 0 \quad (15)$$

Rewriting this equation in the standard unitary form $(T^*)^3 + p(T^*)^2 + qT^* + r = 0$, we have:

$$(T^*)^3 + \frac{\alpha \delta_M a}{\gamma \delta_N b} \left(1 - \frac{\beta}{\gamma} - \frac{\gamma \delta_N}{\alpha \delta_M} \right) (T^*)^2 + \frac{\alpha \delta_M a}{\gamma \delta_N b} \left(2 \frac{\beta}{\gamma} - 1 \right) T^* - \frac{\alpha \delta_M a \beta}{\gamma \delta_N b \gamma} = 0 \quad (16)$$

Notice that the coefficients of this cubic equation depend only on the 3 *non dimensional* parameters $A = a/b$, $B = \beta/\gamma$, and $C = \frac{\gamma \delta_N}{\alpha \delta_M}$.

Remark 2.1 The parameters A , B , and C have the following biological interpretations:

- $A = a/b$. This parameter represents the ratio of the inflammatory effect of the neutrophils w.r.t. the anti-inflammatory effect of the macrophages.
- $B = \beta/\gamma$. This parameter represents the ratio of the increasing rate of the Momac to the decreasing rate of the macrophage.
- $C = \frac{\gamma \delta_N}{\alpha \delta_M} = \left[\frac{\gamma}{\delta_M} : \frac{\alpha}{\delta_N} \right]$. This parameter represents the ratio of the macrophages vs the neutrophils effects in the disease progression.

Using these parameters, the cubic equation in T^* becomes:

$$(T^*)^3 + \frac{A}{C} (1 - B - C) (T^*)^2 + \frac{A}{C} (2B - 1) T^* - \frac{AB}{C} = 0 \quad (17)$$

Remark 2.2 There is an obvious solution in the particular case $A = a/b = 1$, namely $T^* = 1$. This solution corresponds to the disease-free equilibrium point $S^* = 0, N^* = 1, M^* = 1, T^* = 1$. But this solution is not relevant for the case $S^* \neq 0$ that we are considering here. Note further that if $T^* = 1$ then from Eq. (17) we get

$$1 + \frac{A}{C} (1 - B - C) + \frac{A}{C} (2B - 1) - \frac{AB}{C} = 0$$

so that $A = 1$ hence $T = 1$ is a root of the cubic equation (17) iff $A = 1$. Therefore we can factor the cubic polynomial as follows:

$$(T^* - 1) \left((T^*)^2 + \frac{1}{C} (B - 1) T^* + \frac{B}{C} \right) = 0 \quad (18)$$

- If $B \leq 1$, then the coefficients of the quadratic factor are positive, hence there are no other positive roots.
- If $B > 1$, the quadratic factor has either two positive roots (if the discriminant is positive) or no real roots (if the discriminant is negative). The quadratic factor has discriminant $\Delta = \frac{(B-1)^2 - 4BC}{C^2}$. Thus, we have:

– If $\Delta > 0$, there are two distinct real roots given by:

$$T^* = \frac{\frac{(B-1)}{C} \pm \sqrt{\Delta}}{2}$$

– If $\Delta = 0$, there is one (double) positive root given by:

$$T^* = \frac{B - 1}{2C}$$

– If $\Delta < 0$, there are no real roots.

The sign of the discriminant of the quadratic factor depends on the parameters B and C : $\Delta > 0$ iff $\frac{(B-1)^2}{4B} > C$. Let us plot the curve $\frac{(B-1)^2}{4B}$ in the (B, C) plane (see Figure 3). Looking at the figure 3, we see that the typical situation is when $\Delta < 0$, hence there are no other positive roots apart from $T^* = 1$. But for a careful choice of parameters B and C , we can indeed have three equilibria, see fig. 4.

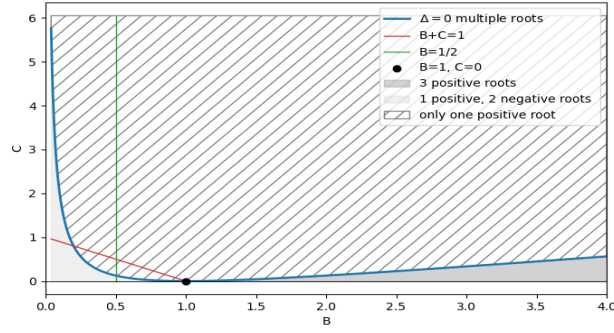


Figure 3: $A = 1$, number of roots.

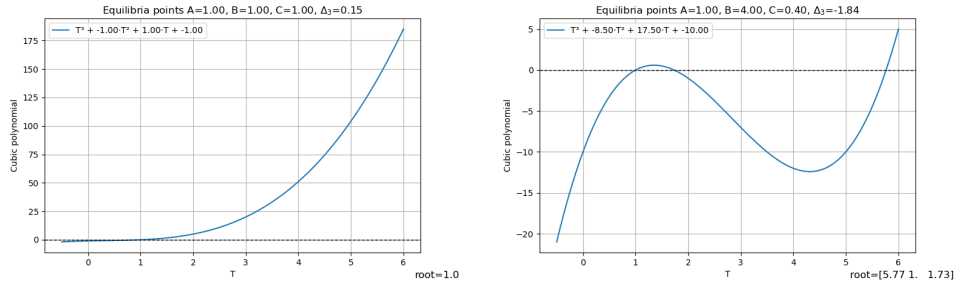


Figure 4: $A = 1$, Left 1 positive root, Right 3 positive roots

Returning to the general case $A \neq 1$, we analyze the number of positive roots of the cubic equation (17). From basic algebra, we can show that there are at most 3 such intersection points, hence at most 3 equilibria points with $S^* \neq 0$. We can be more precise because the biological feasibility imposes the roots to be real nonnegative numbers. Since $\frac{AB}{C} > 0$, there is at least one positive root. But we can be more precise using *Descartes' rule of signs*. It states that the number of positive real roots of a polynomial (counting multiplicities) is equal to the number of sign changes between consecutive nonzero coefficients, or less than that by an even number. In our case, the coefficients of the cubic polynomial in (17) are:

$$1, \quad \frac{A}{C}(1 - B - C), \quad \frac{A}{C}(2B - 1), \quad -\frac{AB}{C}$$

So the number of sign changes depends on the signs of the second and third coefficients. We have the following cases:

- If $B + C > 1$ and $B > 1/2$, then there are 3 sign changes and the cubic has one or three positive roots, which can collapse according to their multiplicities.
- If $B + C < 1$ and $B > 1/2$, then there is one sign change and the cubic has only one positive roots.
- If $B + C < 1$ and $B < 1/2$, then there is one sign changes and the cubic has only one positive root.
- If $B + C > 1$ and $B < 1/2$, then there is one sign change and the cubic has only one positive root.

We can plot some typical cases in the following figure (see Figure 5).

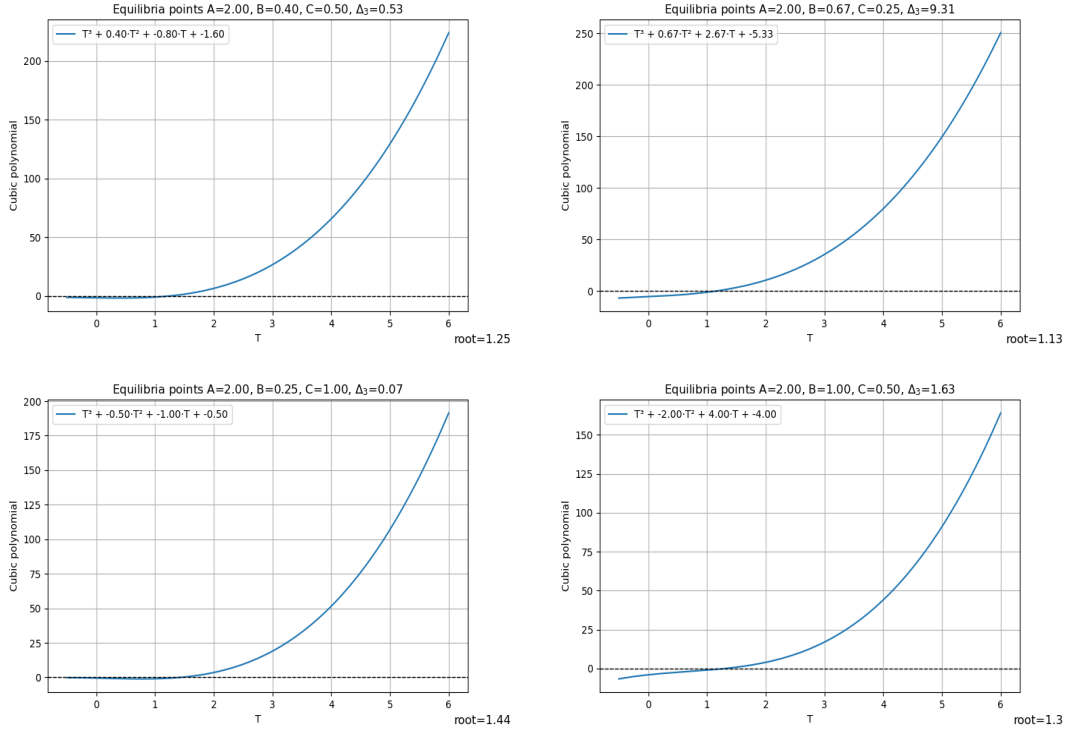


Figure 5: Cubic equation in T^* , left $B < 1/2$, right $B > 1/2$.

We can use Cardano's method to find the roots of the cubic equation (17). We rewrite it in the standard form:

$$T^3 + pT^2 + qT + r = 0$$

where

$$p = \frac{A}{C}(1 - B - C), \quad q = \frac{A}{C}(2B - 1), \quad r = -\frac{AB}{C}$$

We perform the change of variable $T = x - p/3$ to eliminate the quadratic term and obtain the depressed cubic equation:

$$x^3 + \mathcal{A}x + \mathcal{B} = 0$$

where

$$\mathcal{A} = q - \frac{p^2}{3}, \quad \mathcal{B} = \frac{2p^3}{27} - \frac{pq}{3} + r$$

The discriminant of this equation is given by:

$$\Delta_3 = \left(\frac{\mathcal{B}}{2}\right)^2 + \left(\frac{\mathcal{A}}{3}\right)^3$$

The nature of the roots depends on the sign of the discriminant Δ :

- If $\Delta_3 > 0$, there is one real root and two complex conjugate roots.
- If $\Delta_3 = 0$, all roots are real and at least two are equal.
- If $\Delta_3 < 0$, all roots are real and unequal.

Hence we can characterize the case when we have three distinct real roots by the condition $\Delta_3 < 0$, which is equivalent to:

$$\left(\frac{\mathcal{A}}{3}\right)^3 < -\left(\frac{\mathcal{B}}{2}\right)^2$$

This condition can be rewritten in terms of the original parameters using computer algebra as follows:

$$\left(\frac{\mathcal{A}}{3}\right)^3 < -\left(\frac{\mathcal{B}}{2}\right)^2$$

where

$$\begin{aligned}\mathcal{A} &= \frac{A}{3C^2} \left(-A(B+C-1)^2 + 3C(2B-1) \right) \\ \mathcal{B} &= \frac{A}{27C^3} \left(-2A^2(B+C-1)^3 + 9AC(2B-1)(B+C-1) - 27BC^2 \right) \\ \Delta_3 &= \frac{A^2}{2916C^6} \left(-4A(A(B+C-1)^2 - 3C(2B-1))^3 + (2A^2(B+C-1)^3 - 9AC(2B-1)(B+C-1) + 27BC^2)^2 \right)\end{aligned}$$

The real roots can be found using Cardano's formulas. If $\Delta_3 \geq 0$, the real root is given by:

$$x^* = \sqrt[3]{-\frac{\mathcal{B}}{2} + \sqrt{\Delta_3}} + \sqrt[3]{-\frac{\mathcal{B}}{2} - \sqrt{\Delta_3}}$$

If $\Delta_3 < 0$, the three real roots are given by:

$$x_k^* = 2\sqrt{-\frac{\mathcal{A}}{3}} \cos \left(\frac{1}{3} \arccos \left(\frac{3\mathcal{B}}{2\mathcal{A}} \sqrt{-\frac{3}{\mathcal{A}}} \right) + \frac{2k\pi}{3} \right), \quad k = 0, 1, 2$$

Here we plot a case with three distinct positive roots (see Figure 6). Namely we take $A = 1$, $B = 7$ and $C = 1$.

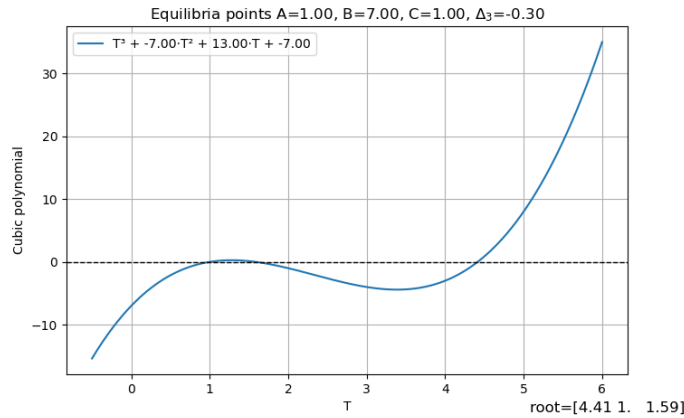


Figure 6: Cubic equation with three distinct real roots.

Remark 2.3 *there is an interesting phenomenon of potential bifurcation when the parameters vary and the number of equilibria points changes. This is a topic for further study. In particular, when the equilibrium point with $S^* \neq 0$ when $S^* \ll 1$ is close to the disease-free equilibrium point $S^* = 0, N^* = 1, M^* = 1, T^* = 1$, we can perform a local stability analysis using perturbation methods. We should analyze this with the tool of catastrophe theory.*

Finally, we can recover the roots T^* of the original cubic equation using the relation $T^* = x^* + p/3$.

2.2 Stability analysis

Stability analysis of the equilibria point (S^*, N^*, M^*, T^*) is done by linearizing the system around this point. The Jacobian matrix of the system is given by:

$$J = \begin{pmatrix} aN - bT & aS & 0 & -bS \\ \alpha N & \alpha S - \delta_N(2N - 1) & 0 & 0 \\ \beta & 0 & -\delta_M(2M - 1) & 0 \\ -\gamma T & 0 & \delta_M(1 - T) & -\gamma S - \delta_M M \end{pmatrix}$$

We analyze first the stability of the disease-free equilibrium point $(S^*, N^*, M^*, T^*) = (0, 1, 1, 1)$. At this point, the Jacobian matrix simplifies to:

$$J|_{(S^*, N^*, M^*, T^*)=(0,1,1,1)} = \begin{pmatrix} a - b & 0 & 0 & 0 \\ \alpha & -\delta_N & 0 & 0 \\ \beta & 0 & -\delta_M & 0 \\ -\gamma & 0 & 0 & -\delta_M \end{pmatrix}$$

We notice immediately that the eigenvalues of this matrix are $a - b$, $-\delta_N$ and $-\delta_M$. The last two are negative since $\delta_N, \delta_M > 0$. The remaining eigenvalue is $a - b$. Thus the disease-free equilibrium is stable if $a < b$ and unstable if $a > b$. This means that if the inflammatory effect of Neutrophiles cells controlled by paramater a is less than the curing effect of the SL-TRM macrophages controlled by parameter b , the system will return to the disease-free state. Conversely, if $a > b$, the system may move away from this equilibrium, potentially leading to a chronic inflammatory state. The stability of other equilibria points can be analyzed similarly by evaluating the Jacobian matrix at those points and determining the eigenvalues.

For $S^* = 0, N^* = 0, M^* = 0, T^* = R$, with arbitrary R , the Jacobian matrix becomes:

$$J|_{(S^*, N^*, M^*, T^*)=(0,0,0,R)} = \begin{pmatrix} -bR & 0 & 0 & 0 \\ 0 & \delta_N & 0 & 0 \\ \beta & 0 & \delta_M & 0 \\ -\gamma R & 0 & \delta_M(1 - R) & 0 \end{pmatrix}$$

The eigenvalues of this matrix are $-bR$, δ_N , δ_M , and 0. Since δ_N and δ_M are positive, this equilibrium point is unstable. Even if the immune signal will decrease exponentially fast, the populations of neutrophiles and momac will grow out of bound. This unstable equilibrium will not be observed in practice.

For $S^* = 0, N^* = 0, M^* = 1, T^* = 1$, the Jacobian matrix is:

$$J|_{(S^*, N^*, M^*, T^*)=(0,0,1,1)} = \begin{pmatrix} -b & 0 & 0 & 0 \\ 0 & \delta_N & 0 & 0 \\ \beta & 0 & -\delta_M & 0 \\ -\gamma & 0 & 0 & -\delta_M \end{pmatrix}$$

The eigenvalues of this matrix are $-b$, δ_N , $-\delta_M$, and $-\gamma$. Since there is a positive eigenvalue, this equilibrium point is unstable. Neutrophiles can grow out of bound. This equilibrium will not be observed in practice neither.

For $S^* = 0, N^* = 1, M^* = 0, T^* = R$, with arbitrary R , the Jacobian matrix is

$$J|_{(S^*, N^*, M^*, T^*)=(0,1,0,R)} = \begin{pmatrix} a - bR & 0 & 0 & 0 \\ \alpha & -\delta_N & 0 & 0 \\ \beta & 0 & \delta_M & 0 \\ -\gamma R & 0 & \delta_M(1 - R) & 0 \end{pmatrix}$$

There is a positive eigenvalue δ_M , hence this equilibrium point is unstable. Momac can grow out of bound. This equilibrium will not be observed in practice neither.

To Do List

- Is there a way to reduce the system introducing new nonlinear variables, taking advantage of the 3 adimensional parameters?
- Recompute equilibria and simplify algebra
- Write Python code to compute equilibria numerically
- perform stability analysis of non homeostasic equilibria (chronic inflammation)
- Perform bifurcation analysis and plot results
- investigate possibility of limit cycles
- is there hysteresis phenomenon?
- Plot 4d phase space trajectories (projections)
- Run numerical simulations with realistic rates
- Discuss model assumptions/results with immunology collaborators

3 Numerical simulations

Here we reproduce one typical simulations of the model using the following parameters: $\alpha = 1, \beta = 1, \gamma = 1, \delta_N = 1, \delta_M = 1, a = 0.35, b = 1$, duration=20 days and initial conditions $S(0) = 1, N(0) = 1, M(0) = 1, T(0) = 1$. We can plot a phase space trajectory in the (S, N, M, T) 4D space as follows:

We need to have realistic values for the parameters to feed the model.

The Python Jupyter notebook used to perform the simulations is available at <https://github.com/azerad/immunomath>.

3.1 Bifurcation analysis

We perform a bifurcation analysis with respect to the *non dimensional* parameters $A = a/b, B = \alpha/\beta, C = \gamma\delta_M/\alpha\delta_M$. We vary A, B, C in the interval $[0.1, 2]$. For each triple (A, B, C) , we simulate the system for 20 days and we record the value of S, N, M, T at the final time. The results are shown in the following heatmap.

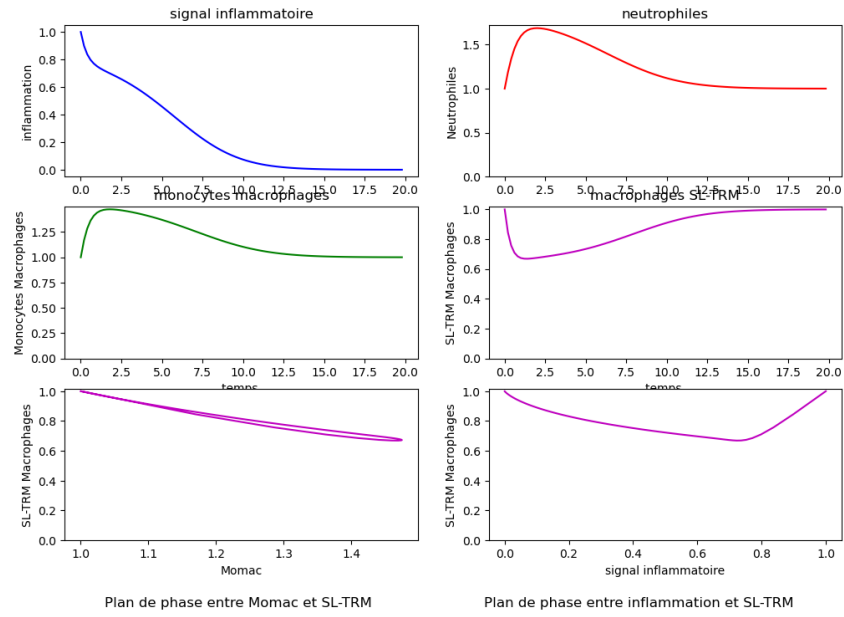


Figure 7:

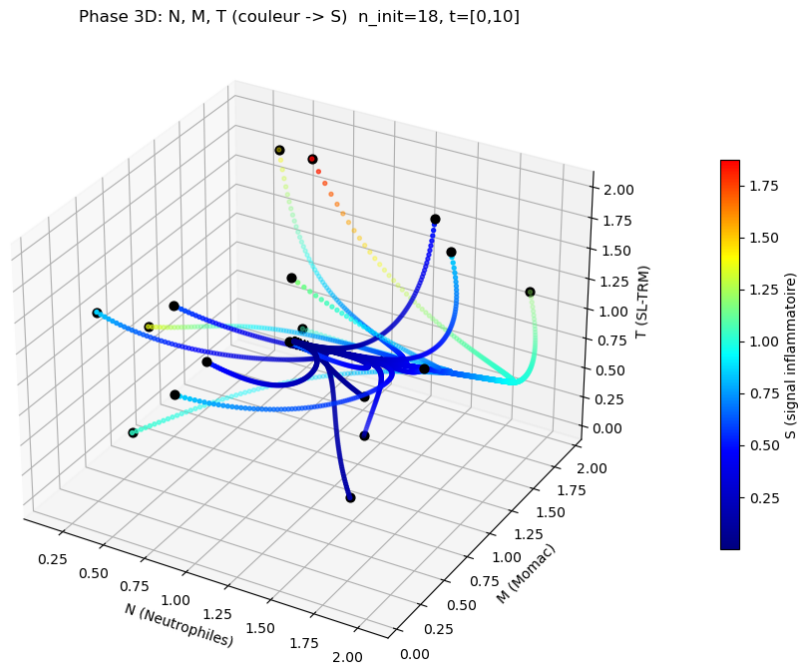


Figure 8: Phase space trajectory in the (S, N, M, T) space.

Acknowledgments

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